The DMAIC Sprint Board

Every Excel statistics function and tool, pinned to the DMAIC phase where a Green or Black Belt reaches for it.

SPRINT RULE
Work top to bottom.
No test before a baseline.

D — DEFINE

Size the problem and find the vital few

13 TOOLS

COUNT THE DEFECTS 5

COUNT(value1, [value2], ...)
The sample size for every other formula

COUNTA(value1, [value2], ...)
Checking for text entries in numeric columns

COUNTBLANK(range)
Missing-data audit

COUNTIF(range, criteria)
Defects beyond a spec limit

COUNTIFS(criteria_range1, criteria1, ...)
Contingency tables and DPU by category

STRATIFY AND PARETO 8

GROUPBY(row_fields, values, function, [field_headers], ...) HELPER
Stratify by heat, shift, line or supplier

PIVOTBY(row_fields, col_fields, values, function, ...) HELPER
Side-by-side comparisons such as LPA vs TWA

PERCENTOF(data_subset, data_all) HELPER
Pareto-style contributions

FILTER(array, include, [if_empty]) HELPER
Slicing a subgroup without copying

UNIQUE(array, [by_col], [exactly_once]) HELPER
Listing categories before grouping

SORT(array, [sort_index], [sort_order], [by_col]) HELPER
Ordering data for plots and ranks

SUBTOTAL(function_num, ref1, [ref2], ...) HELPER
Summaries that follow a filter

AGGREGATE(function_num, options, ref1, [ref2], ...) HELPER
Robust summaries on messy data

M — MEASURE

Baseline the process and its capability

82 TOOLS

DESCRIBE THE DATA 31

AVERAGE(number1, [number2], ...)
Default centre of a data set

AVERAGEA(value1, [value2], ...)
Only when logical values must count

AVERAGEIF(range, criteria, [average_range])
Mean for one shift, grade or line

AVERAGEIFS(average_range, criteria_range1, criteria1, ...)
Mean by heat and orientation together

MEDIAN(number1, [number2], ...)
Skewed data such as delays and cycle times

MODE.SNGL(number1, [number2], ...)
Discrete readings with repeats

MODE.MULT(number1, [number2], ...)
Data with two or more peaks

GEOMEAN(number1, [number2], ...)
Growth rates, ratios, log-normal data

HARMEAN(number1, [number2], ...)
Averaging rates such as units per hour

TRIMMEAN(array, percent)
A robust mean when outliers are present

STDEV.S(number1, [number2], ...)
Almost every quality calculation on sample data

STDEV.P(number1, [number2], ...)
Only when you hold the whole population

STDEVA(value1, [value2], ...)
Rarely; text becomes 0 and distorts results

STDEVPA(value1, [value2], ...)
Rarely; same caution as STDEVA

VAR.S(number1, [number2], ...)
ANOVA, F-tests and pooled variance

VAR.P(number1, [number2], ...)
Complete populations only

VARA(value1, [value2], ...)
Rarely

VARPA(value1, [value2], ...)
Rarely

AVEDEV(number1, [number2], ...)
A spread summary that resists outliers

DEVSQ(number1, [number2], ...)
Building ANOVA sums of squares by hand

SKEW(number1, [number2], ...)
Quick normality and symmetry screen

SKEW.P(number1, [number2], ...)
Skewness of a complete population

KURT(number1, [number2], ...)
Screening for heavy or light tails

MAX(number1, [number2], ...)
Range and worst-case checks

MAXA(value1, [value2], ...)
Rarely

MAXIFS(max_range, criteria_range1, criteria1, ...)
Worst reading per grade or line

MIN(number1, [number2], ...)
Range and minimum-spec checks

MINA(value1, [value2], ...)
Rarely

MINIFS(min_range, criteria_range1, criteria1, ...)
Lowest toughness per heat

LARGE(array, k)
Top-k values and order statistics

SMALL(array, k)
Bottom-k values and order statistics

TURN DATA INTO ODDS 41

NORM.DIST(x, mean, standard_dev, cumulative)
Share of output below a limit, ppm out of spec

NORM.INV(probability, mean, standard_dev)
Percentile limits; simulation with RAND()

NORM.S.DIST(z, cumulative)
p-values for z statistics

NORM.S.INV(probability)
Critical z, sample size, sigma level

GAUSS(z)
Quick area from the centre

PHI(x)
Plotting the bell curve

T.DIST(x, deg_freedom, cumulative)
Lower-tailed t tests

T.DIST.RT(x, deg_freedom)
Upper-tailed t tests

T.DIST.2T(x, deg_freedom)
Two-sided p-value for a t statistic

T.INV(probability, deg_freedom)
One-sided critical values; negative for small p

T.INV.2T(probability, deg_freedom)
Confidence intervals and two-sided tests

CHISQ.DIST(x, deg_freedom, cumulative)
Lower tail of a variance test

CHISQ.DIST.RT(x, deg_freedom)
p-value for chi-square tests with any df

CHISQ.INV(probability, deg_freedom)
Upper limit of a variance confidence interval

CHISQ.INV.RT(probability, deg_freedom)
Chi-square table lookups; variance intervals

F.DIST(x, deg_freedom1, deg_freedom2, cumulative)
Lower tail of a variance-ratio test

F.DIST.RT(x, deg_freedom1, deg_freedom2)
ANOVA and regression p-values

F.INV(probability, deg_freedom1, deg_freedom2)
Lower limit of a variance-ratio interval

F.INV.RT(probability, deg_freedom1, deg_freedom2)
F tables for ANOVA

BINOM.DIST(number_s, trials, probability_s, cumulative)
Defectives in a sample, pass or fail data

BINOM.DIST.RANGE(trials, probability_s, number_s, [number_s2])
Acceptance-sampling ranges in one step

BINOM.INV(trials, probability_s, alpha)
Acceptance numbers and exact critical counts

NEGBINOM.DIST(number_f, number_s, probability_s, cumulative)
Inspections until a set number of finds

HYPGEOM.DIST(sample_s, number_sample, population_s, number_pop, cumulative)
Sampling small lots without replacement

POISSON.DIST(x, mean, cumulative)
Defects per unit, c- and u-chart probabilities

EXPON.DIST(x, lambda, cumulative)
Time between failures, constant failure rate

WEIBULL.DIST(x, alpha, beta, cumulative)
Life data and wear-out reliability

LOGNORM.DIST(x, mean, standard_dev, cumulative)
Repair times, particle sizes, skewed data

LOGNORM.INV(probability, mean, standard_dev)
Percentile limits for skewed data

GAMMA.DIST(x, alpha, beta, cumulative)
Waiting times for several events

GAMMA.INV(probability, alpha, beta)
Percentiles of gamma-shaped data

GAMMA(number)
Building other distributions by hand

GAMMALN(x)
Large factorials without overflow

GAMMALN.PRECISE(x)
Same use, newer algorithm

BETA.DIST(x, alpha, beta, cumulative, [A], [B])
PERT task times, proportions

BETA.INV(probability, alpha, beta, [A], [B])
Exact binomial confidence limits

PERMUT(number, number_chosen)
Counting sequences and orders

PERMUTATIONA(number, number_chosen)
Codes and passwords style counting

PROB(x_range, prob_range, [lower_limit], [upper_limit])
Custom discrete distributions

COMBIN(number, number_chosen) HELPER
Counting samples where order does not matter

COMBINA(number, number_chosen) HELPER
Counting with replacement

POSITION AND SHAPE 10

FREQUENCY(data_array, bins_array)
Histograms and goodness-of-fit counts

QUARTILE.EXC(array, quart)
Box plots that match Minitab

QUARTILE.INC(array, quart)
Matching older Excel workbooks

PERCENTILE.EXC(array, k)
P5 or P95 limits that match Minitab

PERCENTILE.INC(array, k)
Percentiles that include 0 and 1

PERCENTRANK.EXC(array, x, [significance])
Where one value sits in the data

PERCENTRANK.INC(array, x, [significance])
Same, inclusive method

RANK.AVG(number, ref, [order])
Normal probability plots and rank tests

RANK.EQ(number, ref, [order])
League tables and Pareto ordering

STANDARDIZE(x, mean, standard_dev)
Flagging points beyond three standard deviations

A — ANALYZE

Prove which Xs drive Y

39 TOOLS

PUT A ± ON IT 2

CONFIDENCE.T(alpha, standard_dev, size)
Confidence interval for a mean, sigma unknown

CONFIDENCE.NORM(alpha, standard_dev, size)
Confidence interval for a mean, sigma known

TEST THE DIFFERENCE 4

T.TEST(array1, array2, tails, type)
Comparing two means

Z.TEST(array, x, [sigma])
Mean against a target when sigma is known

F.TEST(array1, array2)
Checking variances before choosing T.TEST type

CHISQ.TEST(actual_range, expected_range)
Counts by category; contingency tables

MEASURE THE LINK 7

CORREL(array1, array2)
Strength and direction of a straight-line link

PEARSON(array1, array2)
Legacy sheets; identical result

RSQ(known_y's, known_x's)
Share of variation explained by one x

COVARIANCE.S(array1, array2)
Building r by hand on sample data

COVARIANCE.P(array1, array2)
Matches the ToolPak Covariance tool

FISHER(x)
Confidence interval on r; comparing two r values

FISHERINV(y)
Second half of the r interval

MODEL Y = F(X) 7

LINEST(known_y's, [known_x's], [const], [stats])
Simple or multiple regression with inference

SLOPE(known_y's, known_x's)
Change in y per unit of x

INTERCEPT(known_y's, known_x's)
y when x is zero

STEYX(known_y's, known_x's)
Width of prediction intervals

TREND(known_y's, [known_x's], [new_x's], [const])
Fitted values, residuals, predictions

LOGEST(known_y's, [known_x's], [const], [stats])
Growth or decay curves

GROWTH(known_y's, [known_x's], [new_x's], [const])
Predictions from an exponential model

FULL OUTPUT TABLES (TOOLPAK) 19

Anova: Single Factor · Data › Data Analysis › Anova: Single Factor
Three or more group means

Anova: Two-Factor With Replication · Data › Data Analysis › Anova: Two-Factor With Replication
Two factors with repeats

Anova: Two-Factor Without Replication · Data › Data Analysis › Anova: Two-Factor Without Replication
Randomized block designs

t-Test: Paired Two Sample for Means · Data › Data Analysis › t-Test: Paired Two Sample for Means
Before and after on the same units

t-Test: Two-Sample Assuming Equal Variances · Data › Data Analysis › t-Test: Two-Sample Assuming Equal Variances
Independent groups, equal spread

t-Test: Two-Sample Assuming Unequal Variances · Data › Data Analysis › t-Test: Two-Sample Assuming Unequal Variances
Independent groups, unequal spread

z-Test: Two Sample for Means · Data › Data Analysis › z-Test: Two Sample for Means
Two means with known variances

F-Test Two-Sample for Variances · Data › Data Analysis › F-Test Two-Sample for Variances
Comparing two variances

Descriptive Statistics · Data › Data Analysis › Descriptive Statistics
One-click summary

Regression · Data › Data Analysis › Regression
Full regression output

Correlation · Data › Data Analysis › Correlation
Several variables at once

Covariance · Data › Data Analysis › Covariance
Several variables at once

Histogram · Data › Data Analysis › Histogram
Quick frequency table

Rank and Percentile · Data › Data Analysis › Rank and Percentile
Relative standing table

Random Number Generation · Data › Data Analysis › Random Number Generation
Static simulated data

Sampling · Data › Data Analysis › Sampling
Drawing an audit sample

Moving Average · Data › Data Analysis › Moving Average
Trend smoothing

Exponential Smoothing · Data › Data Analysis › Exponential Smoothing
Level-only forecasts

Fourier Analysis · Data › Data Analysis › Fourier Analysis
Vibration and periodic signals

I — IMPROVE

Find, simulate and prove the fix

18 TOOLS

OPTIMISE AND SIMULATE 18

Goal Seek · Data › What-If Analysis › Goal Seek
Which sigma gives Cpk 1.33?

Data Table (one variable) · Data › What-If Analysis › Data Table (column input cell)
Sensitivity of cost or yield to one input

Data Table (two variables) · Data › What-If Analysis › Data Table (row and column input cells)
Cpk map across mean and sigma

Scenario Manager · Data › What-If Analysis › Scenario Manager
Best, likely and worst cases

Solver (Simplex LP) · Data › Solver › Select a Solving Method: Simplex LP
Blends, product mix, scheduling

Solver (GRG Nonlinear) · Data › Solver › Select a Solving Method: GRG Nonlinear
Response optimization, curve fitting

Solver (Evolutionary) · Data › Solver › Select a Solving Method: Evolutionary
Models with IF, LOOKUP or rounding

Forecast Sheet · Data › Forecast › Forecast Sheet
One-click seasonal forecast

RAND() HELPER
Driving simulations through .INV functions

RANDARRAY([rows], [columns], [min], [max], [integer]) HELPER
Thousands of draws in one formula

RANDBETWEEN(bottom, top) HELPER
Random sampling of rows

SEQUENCE(rows, [columns], [start], [step]) HELPER
Input grids for sensitivity tables

LET(name1, name_value1, calculation_or_name2, ...) HELPER
Readable one-cell simulations

LAMBDA([parameter1, ...], calculation) HELPER
Custom test statistics and models

MAP(array1, lambda_or_array) HELPER
Formula-based data tables that recalculate fast

SUMPRODUCT(array1, [array2], ...) HELPER
Objective and constraint cells for Solver

MMULT(array1, array2) HELPER
Regression by matrices; linear models

MINVERSE(array) HELPER
Normal equations for regression by hand

C — CONTROL

Hold the gain and keep the workbook clean

58 TOOLS

WATCH FOR DRIFT 6

FORECAST.LINEAR(x, known_y's, known_x's)
Straight-line trend forecasts

FORECAST(x, known_y's, known_x's)
Older workbooks only

FORECAST.ETS(target_date, values, timeline, [seasonality], [data_completion], [aggregation])
Time series with trend and seasonality

FORECAST.ETS.CONFINT(target_date, values, timeline, [confidence_level], [seasonality], [data_completion], [aggregation])
Error bands around ETS forecasts

FORECAST.ETS.SEASONALITY(values, timeline, [data_completion], [aggregation])
Confirming the cycle before trusting a forecast

FORECAST.ETS.STAT(values, timeline, statistic_type, [seasonality], [data_completion], [aggregation])
Judging forecast accuracy

REPLACE LEGACY NAMES 39

BETADIST → BETA.DIST
BETAINV → BETA.INV
BINOMDIST → BINOM.DIST
CHIDIST → CHISQ.DIST.RT
CHIINV → CHISQ.INV.RT
CHITEST → CHISQ.TEST
CONFIDENCE → CONFIDENCE.NORM
COVAR → COVARIANCE.P
CRITBINOM → BINOM.INV
EXPONDIST → EXPON.DIST
FDIST → F.DIST.RT
FINV → F.INV.RT
FORECAST → FORECAST.LINEAR
FTEST → F.TEST
GAMMADIST → GAMMA.DIST
GAMMAINV → GAMMA.INV
HYPGEOMDIST → HYPGEOM.DIST
LOGINV → LOGNORM.INV
LOGNORMDIST → LOGNORM.DIST
MODE → MODE.SNGL
NEGBINOMDIST → NEGBINOM.DIST
NORMDIST → NORM.DIST
NORMINV → NORM.INV
NORMSDIST → NORM.S.DIST
NORMSINV → NORM.S.INV
PERCENTILE → PERCENTILE.INC
PERCENTRANK → PERCENTRANK.INC
POISSON → POISSON.DIST
QUARTILE → QUARTILE.INC
RANK → RANK.EQ
STDEV → STDEV.S
STDEVP → STDEV.P
TDIST → T.DIST.RT or T.DIST.2T
TINV → T.INV.2T
TTEST → T.TEST
VAR → VAR.S
VARP → VAR.P
WEIBULL → WEIBULL.DIST
ZTEST → Z.TEST

HAND OFF TO MINITAB OR PYTHON 12

Normality test (Anderson-Darling, Shapiro-Wilk)
In Excel: Normal probability plot with NORM.S.INV and RANK.AVG; check SKEW and KURT

One- and two-proportion tests
In Excel: z statistic with NORM.S.DIST; exact test with BINOM.DIST

Nonparametric tests (sign, Wilcoxon, Mann-Whitney, Kruskal-Wallis)
In Excel: Sign test with BINOM.DIST; rank sums with RANK.AVG

Equal variances, three or more groups (Levene, Bartlett)
In Excel: One-way ANOVA on absolute deviations from each group median

Post-hoc comparisons (Tukey)
In Excel: Fisher LSD with T.INV.2T; no studentized range in Excel

Unbalanced or multi-factor ANOVA and DOE analysis
In Excel: LINEST with coded columns for two-level factorials

Power and sample size
In Excel: Sample-size formula with NORM.S.INV

Process capability
In Excel: Cpk from AVERAGE and STDEV.S, or R-bar over d2

Control charts
In Excel: Limits from AVERAGE and constants, drawn as a line chart

Gage R&R
In Excel: ANOVA method built from DEVSQ

Weibull fitting
In Excel: LINEST on the linearized Weibull plot

Logistic or Poisson regression
In Excel: Solver maximizing the log-likelihood

PY(python_code, return_type) HELPER
scipy.stats and statsmodels for tests Excel lacks

Exam traps: STDEV.S is overall σ (Ppk, not Cpk) · Z.TEST is one-tailed · T.INV is the left tail, use T.INV.2T for ± critical t · I-chart limits use 2.66 × M̄R, not 3 × STDEV.S